

- Q.27 Write down five disadvantages of register ring.
 Q.28 Explain different types of ejection techniques.
 Q.29 Write a short note on size of gate.
 Q.30 Define water connection. Explain its type.
 Q.31 What is balanced runner system? How is it achieved?
 Q.32 Define guide pillar. Write its significance in moulds.
 Q.33 Draw a neat and labelled sketch of two plate mould for suitable article.
 Q.34 Define taper and draft give its significance in injection mould design.
 Q.35 Define O-rings also give their significance in injection mould design.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Define gates and explain different types of gates with neat sketch.
 Q.37 Explain in detail ejector plate assembly with neat sketch.
 Q.38 Write short note on:
 a) Angle lifts splits and its types
 b) Cooling integer type moulds.

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5th Sem / Plastic, Chem Engg. (Spl. Polymer Engg) Subject:- Design of Dies and Moulds- I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The size of the runner depends on the size of the part and its wall thickness.
 a) True
 b) depends on both
 c) Does not depends on any of them
 d) false
- Q.2 _____ type of cooling is used in plastic moulds
 a) Air b) Water
 c) Oil d) All of them
- Q.3 The defects of partial filling of a mould cavity or cavities are called _____.
 a) Flash b) Short shot
 c) Shark skin d) None of these
- Q.4 A thin piece of plastic material that typically forms along the parting line of a moulded part is called _____.
 a) Flash b) Gloss
 c) Tail d) None of these

- Q.5 _____ of runner is important for uniform cavity filling
- Fencing
 - Balancing
 - Tapering
 - None of these
- Q.6 In gating system ratio 1:2:4 represents _____.
 - Sprue base: runner area: gate size
 - Runner area: sprue base: gate size
 - sprue base: gate size: runner area
 - None of these
- Q.7 The purpose of gate in moulding is to _____.
 - To deliver molten plastic into the mould
 - Act as reservoir for molten material
 - Vent out trapped gases
 - Provide shape to the cavity
- Q.8 EDM stands for:
 - Electronic differential machine
 - Electrical discharge machine
 - Engineering down mould
 - None of them
- Q.9 The final portion of the die flow channel just before exiting the die, where the channel cross section is constant, is known as _____.
 - Die swell
 - Die land
 - Die gap
 - All of these
- Q.10 Function of sprue puller is to _____.
 - Eject component
 - To fill gap
 - To remove sprue from its bushing
 - none of these

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The purpose of any machine tool is to remove.
 Q.12 Give function of sprue
 Q.13 Name the mould in which ejector grid is used.
 Q.14 Give two functions of gates.
 Q.15 Name two different types of cooling used in plastic moulds.
 Q.16 The actuator rod is passing through the centre of the _____.
 Q.17 _____ is provided to remove excess air from the mold.
 Q.18 Size of the gate should be kept _____ for good design.
 Q.19 Write function of Bolster plate
 Q.20 Two plate moulds is made up of only two plates. (T/F)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on hydraulic actuation system.
 Q.22 Explain the application of sprue bush.
 Q.23 Write a short note on indirect bolting method.
 Q.24 Explain deciding factors for runner size.
 Q.25 Write short note on importance of cooling.
 Q.26 Differentiate between hot runner mould and two plate mould.

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